Project: Worked examples
Title: Plane wall heat transfer (layers and films)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Steady heat flow through a composite plane wall between two fluids, or between two known surface temperatures: each layer's conduction resistance, each film's convection resistance, the overall U-value, the heat flow and flux, and every surface and interface temperature.

WHAT IT CALCULATES	RELATION
Resistance of each surface film	R = 1/(h·A)
Resistance of each conducting layer	R_i = t_i/(k_i·A)
Total resistance, hot side first	R_tot = Σ R
Heat flow through the wall	Q = (T_1 − T_2)/R_tot
Heat flux and overall U-value	q = Q/A; U = 1/(R_tot·A)
Surface and interface temperatures, stepping down from T_1 across each resistance in turn	T_next = T_previous − Q·R

Basis: The plane wall: thermal resistance and the composite wall[1](§3.1); Generalized thermal resistance networks (multilayer plane walls)[2](§3-3); Thermal resistance of building components (method)[3](§6).

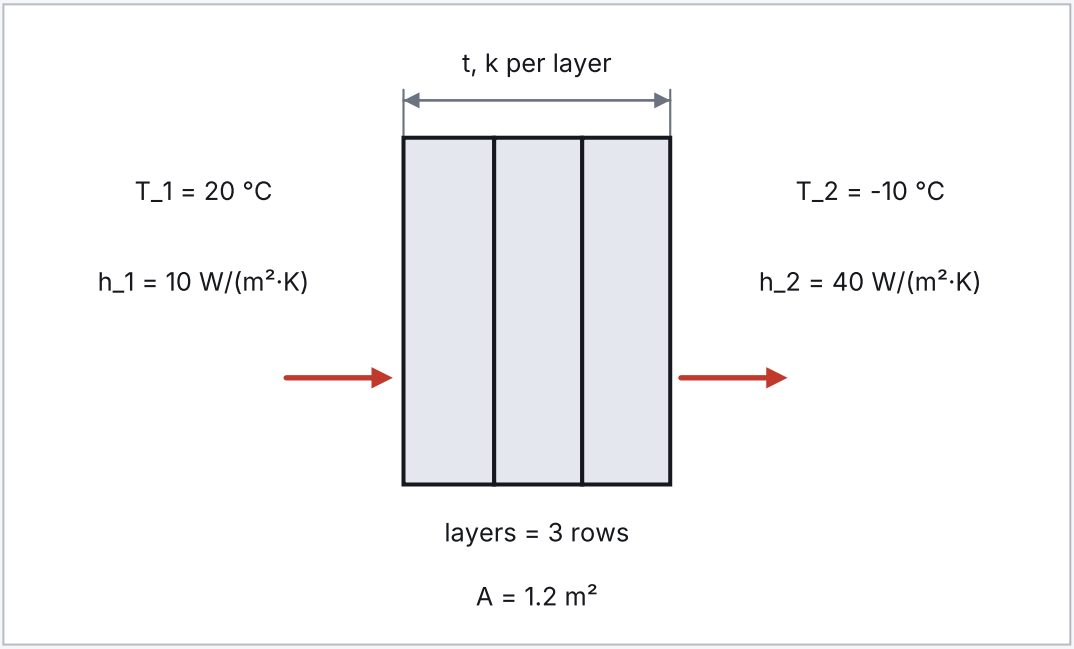


Figure 1 — A composite plane wall, side 1 left: a fluid film, the layers as listed (three drawn), a film on side 2. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
T_1	Side 1 temperature (the fluid's, or the surface's when no film is given)	20	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h_1	Side 1 film coefficient (omit when T_1 is the surface temperature)	10	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)
layers	Conducting layers, side 1 first	3 row(s)		drawing: DRG-GL-802 rev A	any
h_2	Side 2 film coefficient (omit when T_2 is the surface temperature)	40	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)
T_2	Side 2 temperature (the fluid's, or the surface's when no film is given)	-10	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C
A	Wall area normal to the heat flow	1.2	m²	drawing: DRG-GL-802 rev A	0.001 … 10000 m²

TABLE ROW	VALUES
layers[1]	name = glass; t = 4 mm; k = 0.78 W/(m*K)
layers[2]	name = air gap; t = 10 mm; k = 0.026 W/(m*K)
layers[3]	name = glass; t = 4 mm; k = 0.78 W/(m*K)

3. Assumptions

A1. Steady, one-dimensional conduction through plane layers of constant conductivity, with no internal heat generation. (Thermal-circuit idealisation: equal layer areas, no edge losses.)
A2. Layers are in perfect thermal contact; each film coefficient is constant over its surface. (A contact resistance can be entered as a thin layer of equivalent thickness and conductivity.)

4. Working

[6] R_h1 = 1 / (h_1 · A) [1](§3.1)
= 1 / (10 W/(m²·K) × 1.2 m²)
= 0.083333 K/W
Side 1 film resistance

[7] R_1 = t_1 / (k_1 · A) [1](§3.1)
= 4 mm / (0.78 W/(m·K) × 1.2 m²)
= 0.0042735 K/W
Conduction resistance of glass

[8] R_2 = t_2 / (k_2 · A) [1](§3.1)
= 10 mm / (0.026 W/(m·K) × 1.2 m²)
= 0.32051 K/W
Conduction resistance of air gap

[9] R_3 = t_3 / (k_3 · A) [1](§3.1)
= 4 mm / (0.78 W/(m·K) × 1.2 m²)
= 0.0042735 K/W
Conduction resistance of glass

[10] R_h2 = 1 / (h_2 · A) [1](§3.1)
= 1 / (40 W/(m²·K) × 1.2 m²)
= 0.020833 K/W
Side 2 film resistance

[11] R_tot = Σ R [2](§3-3)
= 0.083333 + 0.0042735 + 0.32051 + 0.0042735 + 0.020833 K/W
= 0.43323 K/W

Total thermal resistance

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[12] Q = (T_1 - T_2) / R_tot
= (20 degC - (-10 degC)) / 0.43323 K/W
= 69.248 W
```

Heat flow from side 1 to side 2

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[13] q = Q / A
= 69.248 W / 1.2 m²
= 57.707 W/m²
```

Heat flux

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[14] U = 1 / (R_tot * A)
= 1 / (0.43323 K/W × 1.2 m²)
= 1.9236 W/(m²·K)
```

Overall heat transfer coefficient

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[15] T_s0 = T_1 - Q · R_h1
= 20 °C - 69.248 W × 0.083333 K/W
= 14.229 degC
```

Side 1 surface

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[16] T_s1 = T_s0 - Q · R_1
= 14.229 °C - 69.248 W × 0.0042735 K/W
= 13.933 degC
```

Interface 1

```
[17] T_s2 = T_s1 - Q · R_2
= 13.933 °C - 69.248 W × 0.32051 K/W
= -8.2614 degC
```

Interface 2

```
[18] T_s3 = T_s2 - Q · R_3
= -8.2614 °C - 69.248 W × 0.0042735 K/W
= -8.5573 degC
```

Side 2 surface

5. Results

SYMBOL	VALUE	DESCRIPTION
Q	69.248 W	Heat flow
q	57.707 W/m²	Heat flux
U	1.9236 W/(m²·K)	Overall heat transfer coefficient
R_tot	0.43323 K/W	Total thermal resistance
R_h1	0.083333 K/W	Side 1 film resistance
R_1	0.0042735 K/W	Layer 1 resistance
R_2	0.32051 K/W	Layer 2 resistance
R_3	0.0042735 K/W	Layer 3 resistance
R_h2	0.020833 K/W	Side 2 film resistance
T_s0	14.229 degC	Side 1 surface temperature
T_s1	13.933 degC	Interface 1 temperature
T_s2	-8.2614 degC	Interface 2 temperature
T_s3	-8.5573 degC	Side 2 surface temperature

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

No warnings.

8. Verification

```
Template bs.plane-wall@1.0.1 Content hash d3270684be534c78 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
2. Çengel YA, Ghajar AJ. Heat and Mass Transfer: Fundamentals and Applications. 5th ed. McGraw-Hill Education; 2015.
3. British Standards Institution. Building Components and Building Elements—Thermal Resistance and Thermal Transmittance—Calculation Methods. BS EN ISO 6946. BSI.